1 Booking of Quantities

Data collection in different quantity types and quantity accounts

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts.

The following **quantity accounts** are supported:

- Yield
- Scrap
- Rework
- Open quantity (problem quantity)

The following **quantity types** are supported with each quantity account:

- Primary quantity
- Secondary quantity
- Tertiary quantity
- Base quantity

The automatic collection of quantities on the terminal always refers to the primary quantity.

Conversion to alternative quantity units

Bookings for alternative quantity accounts can be performed as follows:

- direct (manual) input
- conversion
    - from other quantity types with manual input
    - from primary quantity with automatically collected quantities

Direct (manual) input

If a quantity is directly (manually) entered in a quantity unit of a quantity type, an automatic conversion is not performed.

Conversion from other quantity types

If alternative quantity accounts are not collected, the server converts the quantities to alternative accounts using the conversion factors or quantity units, which are configured in the master data of the machine/workplace on the console.

In general, conversion first takes place into the base quantity unit (unless this one is recorded manually) and from the base quantity unit into the alternative unit (unless this one is recorded manually).

Identical quantity units

If quantity units are identical they are converted via numerator and denominator in the master data of machines/workplaces. If numerator and/or denominator = 0, then the quantities are taken over 1 to 1 without being converted.

Different quantity units

If quantity units are different, conversion is performed in the following order of priorities:

- Conversion using the numerator and denominator defined in the workplace/machine master data (always convert into base quantity unit first, then into the alternative unit);
  If numerator and/or denominator = 0

- Conversion using the formulas of the quantity units .

No quantity units

If quantity units are not assigned for the machine, no conversion is performed.

Conversion of quantity 0

A quantity 0 is generally not converted into alternative units, even if a value that is not 0 could be calculated (e.g. using a formula).

Quantity conversion of automatically collected quantities

To convert automatically recorded quantities using formulas, you can use
- fixed factors/values or values based on machines/workplaces (user fields);
- data that is specific to the operation, such as length, width, weight per piece, etc.

You use the operation logged on the longest to identify the OP-specific data. Consequently, the following (logical) restriction arises for operations that are logged on at the same time when it comes to quantity conversion:
- the operations must produce the same material;
- the operations must have the same default data (length, width, weight per piece).

Any further requirements must be taken into account as part of customer projects.

Basis for HYDRA-MDE quantity conversion

In the *Workplace/machine configuration*, tab *Configurations > Quantities*, you can use the configuration option *Basis for HYDRA-MDE quantity conversion* to use the configured quantity conversion of the running operations also for the machine. This option ensures a correct calculation of quantities even if more than one operation is active :
M – conversion factors of the workplace (APZ) [default]
A – conversion factors of the OP if logged on, otherwise workplace

Notes

If configuration A and different quantity units of operations are used, the quantities are accumulated and booked to the machine accounts without reference to the units.

When you edit postings, the MDE-related quantity conversion using the operation data or the machine/workplace values (user fields) is not supported. In this case, do not enable the option *Convert quantities* in the editing dialog of the maintenance of postings.

Display of alternative quantity units on the terminal

The (manual) collection and display of alternative quantity units is only possible on Windows terminals (requires customization). Note: The terminal itself does not perform any local conversion into alternative quantity units; the quantities are only displayed when data is reloaded from the HYDRA server.

Automatic collection of quantities

You configure the counters for each machine. The following options are available:

- Posting of yield, scrap, rework, open quantity, no posting
- Posting of cycles (strokes)
- The quantity is calculated using partitioning (parts produced per cycle) and/or pulse factor
- Cycle monitoring
- Reason (e.g. scrap reason)
- Offset against “quantity account"
  e.g. scrap is deducted from yield

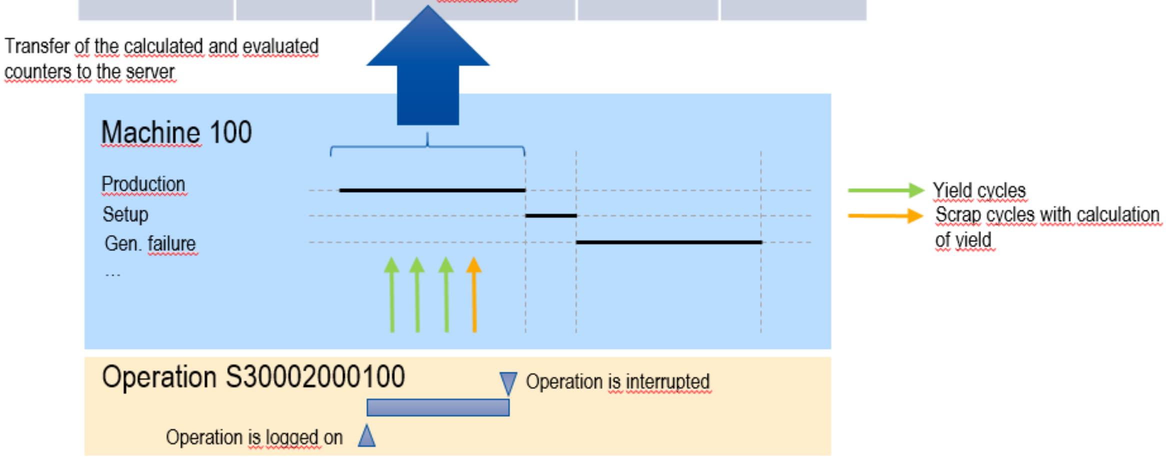
Quantities that are issued by automatically recorded counters are always posted in primary quantity. The conversion into other quantity types is described in the previous section. The HYDRA server can use different quantity accounts to calculate a quantity (e.g. to record the total quantity).

If the cycles are collected, not every single cycle is transferred to the server, but the collected and also the calculated cycles and the evaluated counters are cyclically transferred to the server. The values transferred are then integrated into the events or into the calculated quantities of the machine-related postings (MDE log records).

Example 1:
A yield counter and a scrap counter are defined for the machine. The scrap collected is offset against the yield collected.
The events and postings include the offset and calculated quantities. Not for each cycle, a posting is created.

Postings for orders

Order/OP	Workplace	Record type	Yield (P)	Scrap (P)
S30002000100	M100	Part quantity posting	2	0
S30002000100	M100	Part quantity posting	0	1
S30002000100	M100	Order interruption	2	1

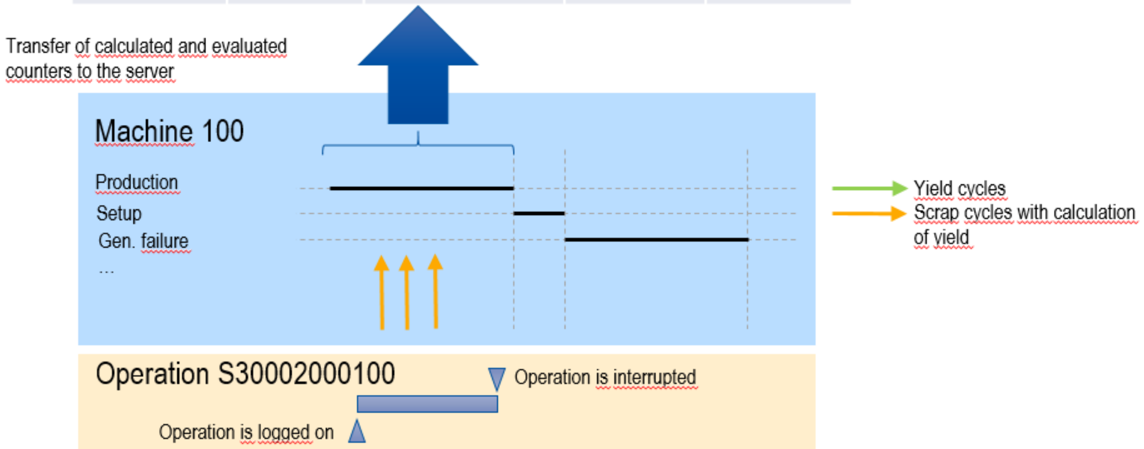


Example 2:

A defined calculation of a counter reading can also result in negative quantities in the posting records:

Postings for orders

Order/OP	Workplace	Record type	Yield (P)	Scrap (P)
S30002000100	M100	Part quantity posting	-3	3
S30002000100	M100	Order interruption	-3	3



Counters that are configured with the "no posting" option do not post any quantities to a quantity account. Using these counters, the following use cases can be integrated:

- Cycle monitoring only

- Cycle monitoring and posting as cycles (strokes)

- Posting as cycles (strokes) without cycle monitoring

Quantity calculation and parallel make-to-order production

With operations that are logged on at the same time, quantities are posted with respect to the order according to the relevant specifications (partitioning, pulse factor) of the operation.

This specific calculation of quantities is performed with all quantity accounts that are recorded automatically (yield, scrap, rework, open quantity). Counter pulses or quantities resulting from this calculation are generally posted onto *all* OPs that are logged on (according to the configuration: with activated partitioning/pulse factor or not).

Specifications are identified as follows:

Partitioning/cavity (TLG)

Partitioning of the machine (parts per cycle): $(TLG_{OP1} + TLG_{OP2} + TLG_{OPn}) * TLG_{Machine}$

If a partitioning is specified for a machine, the partitioning of the operations and the machine partitioning are multiplied.

If you interrupt/log off an operation, the partitioning of the machine is updated, i.e. the total of the remaining operations is recalculated.

If the last operation is interrupted or logged off, the partitioning is reset to 1 and multiplied by the machine-specific partitioning.

Partitioning of the operation: $TLG_{OP} * TLG_{Machine}$

Changing the partitioning on the terminal

If the partitioning of the operation is changed, the partitioning of the machine and of the operation is changed/updated.

Pulse factor (IMPFAKT)

Pulse factor of the machine = minimum($IMPFAKT_{OP1}$, $IMPFAKT_{OP2}$,… $IMPFAKT_{OPn}$) * $IMPFAKT_{Machine}$

Pulse factor of the OP = minimum($IMPFAKT_{OP1}$, $IMPFAKT_{OP2}$,… $IMPFAKT_{OPn}$) * $IMPFAKT_{Machine}$

Note:
The same pulse factor applies for all active operations. Therefore, you must ensure that parallel operations get the same default pulse factor.
This means: The same pulse factor is used for all operations when quantities are calculated.

Quantity calculation on the basis of partitioning and pulse factor

Quantity of the machine = <number of cycles> * partitioning of the machine / pulse factor of the machine

Quantity of the operation = <number of cycles> * partitioning of the operation / pulse factor of the operation

Note:
The pulse factor is calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

Display on terminal

On the terminal, the field **Partitioning** shows the factor that is relevant for the quantity calculation of the machine. It is

partitioning of the machine / pulse factor of the machine.

As described above, the different default values of the machine (e.g. machine-specific partitioning) and of the active operations are used to calculate this factor.

On the terminal, the field **Target cycle** shows the relevant current target cycle (the cycle extension is not integrated). This is max($SZY_{OP1..n}$) for OPs that are logged on at the same time.

Output "target quantity reached"

The target quantity output of the machine interface, e.g. setting a lamp, is set when the smallest target quantity of a logged on order is reached. If the OP with the smallest target quantity is interrupted or finished, the next OP with the smallest target quantity is used.

Manual collection of quantities

You can also use the configurations below to book manually recorded quantities:

- Offset quantities (accounts) against other quantity accounts (*Allocation with* option)
  e.g. deducting manually recorded scrap from yield

- Post manual quantities as cycles

Offsetting against another quantity account

You can offset automatically and manually recorded quantities against another quantity account, e.g. you can deduct the manually recorded scrap from the yield.

You can use the *Allocation with* option for

- automatically recorded quantities in the counter configuration

- manually recorded quantities in the machine configuration.

If you use these options, bookings (BDE log records, MDE log records) with negative quantities or negative order quantities can result.